Knotted Coil Designer

A browser tool for designing MRI receive phased arrays made of **knotted wire loops** (torus knots T(p,q), with 1–8 lobes) on a helmet former. It generates single elements, “blooms” them, and **automatically arranges N elements to reduce coupling between neighbours (geometric decoupling)**. It exports wire paths, coupling matrices and meshes.

> **Status: exploratory research software.** Every electromagnetic output is a *quasi-static geometric estimate*: it assumes low frequency and models the wires as thin current paths. It ignores capacitors, the head/body load and wave effects. Confirm any design with a full electromagnetic (full-wave) simulation or with bench S21, the coupling between two coils measured on a network analyser, before building.

How to run

- **Easiest:** double-click `app/index.html`. It needs no install and no internet; the Three.js 3D library is stored in `vendor/`.
  - *Not yet tested:* opening by double-click. It was tested served from a local web server.
  - If the array solver doesn’t respond when opened by double-click, use the built-in local server (tested). Run `node tools/serve.js` in this folder, then open http://localhost:8765/app/index.html.
- **Validation suite** (needs Node.js; takes about 2 minutes):

```
node tests/run.js
```

This writes `validation/validation_report.html`. The app’s header shows the latest verdicts.

What's where

Path	Contents
`DEVELOPER_GUIDE.md`	**Start here if you want to extend the tool:** architecture, how to add knot families, rules
`CLAUDE.md`	Instructions for AI coding assistants working on this project
`docs/REQUEST_LOG.md`	The owner's design requests, quoted, and what each produced
`docs/research_integrity/`	The research-integrity rules this project follows (Reza Farivar's framework, DRAFT v1.0)
`sources/`	Archived primary sources for verified citations
`NOTICE.txt`	Copyright, sharing terms, status
`Design.md`	Frozen design v1.5. The changelog records every version: v1.0 pre-registered V1–V6; V7 and V8 were added before they ran
`DECISIONS.md`	Dated decision log, including validation failures and open questions for Reza
`PARAMETERS.md`	Every parameter with its origin tag (LIT / CONV / JUDG / TUNED / DERIVED)
`core/`	Geometry and EM package: `knot.js` , `inductance.js` , `surface.js` , `layout.js` , `optimise.js`
`app/index.html`	Browser UI: Element, Pair explorer, Array on helmet
`tests/run.js`	Pre-registered validation checks plus labelled post-hoc checks
`tests/compare_worstpair.js`	Pre-registered solver comparison
`tools/serve.js` , `tools/build_dist.js`	Local server; builds `dist/` (a flat, hostable copy of the app)
`validation/`	Generated validation reports
`vendor/`	Three.js 0.149.0 (MIT licence, `vendor/three-LICENSE.txt`)

Validation status (run 6, 2026-10-20, Design v1.5)

Check	Verdict
V1 Maxwell coaxial mutual inductance	PASS (worst error 0.005%)
V2 Loop self-inductance	PASS (worst error 0.66%)
V3 Coplanar-circle null spacing	NOT EVALUABLE (reference unverified; computed 0.760 × diameter)
V4 Convergence	PASS
V5 Symmetry	PASS
V6 Topology across bloom (β 0.02–3.0)	**FAIL** only below the wire-dependent bloom floor (β = 0.02, 0.05). It is a physical limit, shown in the app
V7 Topology across petal sharpness	PASS (all 360 points). Earlier failures are kept on record: run 4 failed under the v1.3 definition, and run 5 failed at 1 of 360 points from a point-count artefact
V8 B1 on-axis loop field	PASS (worst error 0.005%)

Every failing run is kept in the git history; see DECISIONS.md.

B1 quick look (Array tab): shows the quasi-static Biot–Savart field of the solved wires.

- **Views:** on a placeholder head surface, or as an axial, coronal or sagittal slice.
- **Elements:** combined as root-sum-of-squares, or a single element (click its wire).
- **Use it for:** comparing coverage and depth fall-off between designs. It is not an absolute sensitivity prediction, because tissue and wave effects are not modelled.
- **E-fields are not shown:** this model can't produce them reliably. Use a full-wave simulator with the exported wire paths.

Solver comparison (pre-registered): `validation/compare_worstpair.html`. The worst-pair term is on by default. It lowers the worst pair but raises the typical (median) coupling; set "Worst-pair weight" to 0 to turn it off.

References

- **Rosa, E. B. & Grover, F. W. (1912).** Formulas and tables for the calculation of mutual and self-inductance (revised). *Bulletin of the Bureau of Standards* 8(1):1–237, Scientific Paper 169. https://nvlpubs.nist.gov/nistpubs/bulletin/08/nbsbulletinv8n1p1_A2b.pdf
  - p. 6, eq. [1]: Maxwell's coaxial formula, modulus convention. **VERIFIED**.
  - p. 111, eq. [62]: ring self-inductance with surface current. **VERIFIED**.
  - pp. 20–21: Example 1, used for post-hoc check PH1. **VERIFIED**.
- **Roemer, P. B., Edelstein, W. A., Hayes, C. E., Souza, S. P. & Mueller, O. M. (1990).** The NMR phased array. *Magn Reson Med* 16:192–225. doi:10.1002/mrm.1910160203. **UNVERIFIED** (not read).

- **Roemer, P. B. & Edelstein, W. A., US Patent 4,825,162 (1989).** Read via Google Patents OCR text: circular-coil spacing “D approximately equal to 0.75d”. **PARAPHRASE-ONLY.**

Provenance and AI-use statement

This software was written with **Claude Code** (Anthropic’s AI coding assistant, model Claude Opus 5.5) under the direction of **Reza Farivar** (McGill University), who is accountable for it.

- Every source file carries a provenance header.
- Reference formulas were verified against the primary text listed above.
- Validation criteria were pre-registered in `Design.md` (git 39a4e83) before any code was written.
- Failures are recorded in `DECISIONS.md` and in the git history.
- Any use of results in publications must disclose AI assistance as the venue requires (COPE/ICMJE for journals; current Tri-Agency guidance for grants).

Third-party code: Three.js 0.149.0, © three.js authors, MIT licence.